

Technology Stack

Date	29 June 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to build all three web pages — home.html, indexnew.html, resultnew.html. CSS provides dark themed interactive design. JavaScript powers the live HDI score estimator and progress tracker on the form page

2	Backend / Server-Side	Python 3.x, Flask Framework	Flask is a lightweight Python web framework used to handle routing, form data processing, and ML model integration. Python handles all data processing and prediction logic in app.py
3	Database / Data Storage	CSV File (HDI.csv), Pickle File (HDI.pkl)	HDI.csv stores the UNDP training dataset with 191 countries. HDI.pkl is the serialized trained Linear Regression model saved using Python's Pickle module for reuse without retraining
4	Cloud / Hosting / Deployment	Local Server (localhost:5000) via Flask built-in server	The application runs locally on port 5000 using Flask's built-in development server. Can be deployed to cloud platforms like Heroku or Render in future

5	Version Control & CI/CD	GitHub	Project code is maintained on GitHub for version control, team collaboration, and submission of the project link to the SkillWallet platform
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook	Scikit-learn provides Linear Regression algorithm. Pandas handles dataset loading and preprocessing. NumPy handles numerical operations. Matplotlib and Seaborn generate visualizations. Jupyter Notebook is used for model training